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Paleomagnetism of the Morro Rock–Islay Hill Complex as Evidence for Crustal Block Rotations in Central Coastal California

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Paleomagnetic directions obtained from seven hypabyssal intrusive bodies of the late Oligocene Morro Rock–Islay Hill igneous complex of central coastal California display clockwise deflections from the expected mid-Tertiary direction. Both alternating field and thermal demagnetization experiments show these magnetization directions to be stable and essentially free from secondary overprinting. The mean paleomagnetic direction for all sites is rotated clockwise 49°. Moreover, different bodies show different amounts of clockwise rotation, with declinations ranging from 27° to 76°. These results can best be explained by the clockwise rotation of several local crustal blocks. It is suggested that these rotations are the result of right lateral shearing acting on extensionally loosened crustal fragments within a pull-apart basin. This combines two previously suggested ideas, the 'ball bearing' rotation hypothesis of Beck (1976) and the pull-apart basin process discussed by Crowell (1974), both of which appear to be important tectonic processes in the extensively sheared continental margin in California.

INTRODUCTION

Paleomagnetic studies of the western edge of the North American continent have yielded paleomagnetic directions which show considerable divergence from the directions expected for a stable North America [*Cox*, 1957; *Watkins*, 1965; *Symons*, 1971; *Teissere and Beck*, 1973; *Beck*, 1976; *Simpson and Cox*, 1977]. Most of these divergent directions show clockwise deflections of declination and shallower inclinations than expected. The western margin of North America in California exhibits a system of subparallel, en echelon, and branching right slip faults marking a zone of distributed dextral shear. This broad transform zone [*Atwater*, 1970] defines the boundary between the Pacific and North American plates. To explain the paleomagnetic data, *Beck* [1976] proposed that within this zone, rigid microcontinental fragments between faults rotate clockwise and are translated northwestward like ball bearings in response to the dextral shear of the bounding faults (Figure 1). Paleomagnetic results from southern Europe have similarly been interpreted as being due to left lateral shear between the Eurasian and African plates [*Zijderveld and Van der Voo*, 1973].

Although the paleomagnetic results cited above leave little doubt that crustal blocks have undergone large rotations about local vertical axes, there is still some uncertainty about the nature of the tectonic processes involved and the dimensions of the rotated crustal blocks. *Irving* [1964] pointed out that the result of *Cox* [1957] from Eocene volcanics in Oregon was consistent with the proposed Mendocino orocline of *Carey* [1958], which has a length of about 10^3 km. *Watkins* [1965] similarly interpreted paleomagnetic results from the Columbia basalts as being due to a small amount of post-Miocene oroclinal bending. On the other hand, *Simpson and Cox* [1977] concluded that a coastal block in Oregon at least 225 km long, but smaller than Carey's orocline, had rotated by an angle of 60° since the Eocene and suggested that the tectonic mechanism was analogous to back arc spreading. In their model the length of the block is much larger than the thickness of the lithosphere, so the rotated block might legitimately be regarded as a microplate. In presenting his 'ball bearing' hypothesis, *Beck* [1976] did not specify the actual dimensions of the rotating blocks, although he implied that their diameters would be comparable with the spacing between the bounding faults. For closely spaced faults the balls might be quite small, and in the limiting case the ball bearing model would be identical with distributed penetrative shear in a fault zone.

A broad zone of shear can lead to other tectonic processes as well. *Crowell* [1974] has discussed how a zone of branching right slip faults can produce tipped wedge basins and pull-apart basins. The pull-apart process appears to be responsible for the formation of several major sedimentary basins in southern California, including the Los Angeles basin [*Crowell*, 1974] and the Santa Maria–Lompoc basin (C. A. Hall, unpublished manuscript, 1977).

This paper presents yet another case of divergent paleomagnetic directions showing clockwise rotations and discusses them in terms of the concepts of pull-apart basins and ball bearing rotations.

GEOLOGICAL SETTING

The formation sampled in this paleomagnetic study, the Morro Rock–Islay Hill complex, is a line of hypabyssal volcanic plugs, domes, and dikes exposed in the northwest trending Los Osos Valley, San Luis Obispo County, California. The major bodies form a linear chain from Morro Rock, the westernmost member, to Islay Hill, approximately 25 km to the southeast (Figure 2). Each body is approximately 1 km in diameter with the exception of the one larger body, which is exposed as several small outcrops but has been mapped as a single unit by *Jennings* [1958]. Radiometric ages (K-Ar) for the complex range from 22.1 ± 0.9 m.y. to 26.5 ± 0.8 m.y. [*Turner*, 1968], dating the emplacement of these bodies as late Oligocene.

The Morro Rock–Islay Hill complex is intrusive into the melánge and metavolcanics of the Franciscan Formation. The structural incoherency of the Franciscan rocks and the absence of younger bedded deposits at the sampling sites make it virtually impossible to determine how much the intrusive bodies have tilted since cooling, if such changes in attitude have indeed occurred. However, regional fold trends place important limitations on the directions of possible tilting, as will be discussed later.

The complex lies within the large continental sliver between the San Andreas fault and the offshore Hosgri fault, along which a considerable part of the motion between the North

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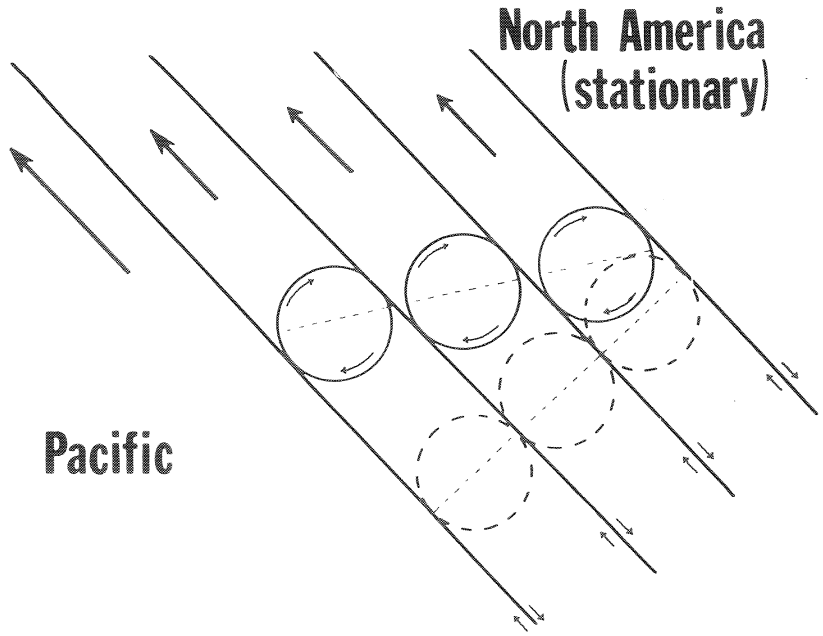


Fig. 1. Cartoon illustrating northwestward translation and clockwise rotation of rigid crustal blocks in a northwestward trending right lateral shear zone. Heavy diagonal lines represent individual faults in a zone of distributed shear. Crustal blocks (represented by circles) can rotate ball bearing fashion or move northwestward without rotation [after *Beck*, 1976].

American and Pacific plates has taken place [*Graham and Dickinson*, 1978]. The bodies of the complex lie along the axis of a valley which is bounded on the northeast by the Cambria and Oceanic faults and on the southwest side by the Edna fault zone (Figure 2). Other faults mapped in the local area are the Rinconada fault, northeast of the Oceanic fault, and the Miguelito fault, south of the Edna fault. The Cambria, Oceanic, and Edna faults converge approximately 25 km southeast of Islay Hill to form the West Huasna fault. All of the above faults have a northwesterly trend with the exception of the Hosgri fault, which has a more northerly trend in this area.

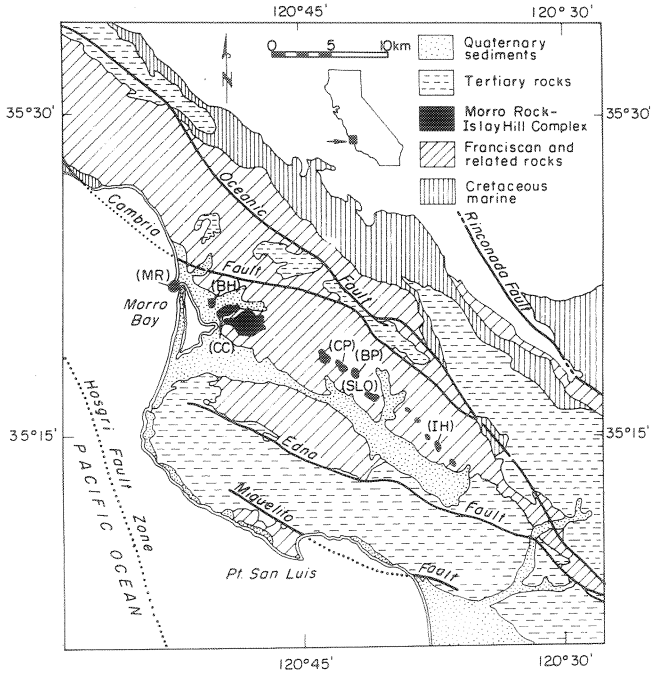


Fig. 2. Generalized geology of the sampling area. Sampled bodies of the Morro Rock–Islay Hill complex are indicated by the letters in parentheses: MR, Morro Rock; BH, Black Hill; CC, Cerro Cabrillo; CP, Chumash Peak; BP, Bishop Peak; SLO, Cerro San Luis Obispo; IH, Islay Hill (modified from Geologic Map of California, San Luis Obispo Sheet [*Jennings*, 1958]; fault locations from *Hall* [1973, 1974] and *Hall and Prior* [1975].

Tertiary rocks on both sides of the valley show late Pliocene or early Pleistocene folding [*Ernst and Hall*, 1974]. The folds are long, narrow, and trend in a northwest-southeast direction, essentially parallel to the local faults, the valley, and the Morro Rock–Islay Hill complex.

The rocks of this complex are porphyritic dacites, and petrographic analysis by *Ernst and Hall* [1974] revealed the following mineralogical characteristics. The phenocrysts comprise half the rock and include zoned intermediate plagioclase, partially resorbed quartz, brown biotite, magnetite, and hornblende. The groundmass contains essentially the same minerals. Magnetite constitutes about 5% of the larger grains and also occurs in the groundmass and as secondary rims around oxidized biotite and hornblende grains. In the present study, samples from one of the intrusive bodies, Islay Hill (IH), contain a significant amount of very fine grained reddish hematite. Demagnetization experiments on specimens from this site revealed coercivities and blocking temperatures which showed that hematite carries a large fraction of the total magnetization. In samples from the other bodies the principal magnetic carrier proved to be fine-grained magnetite.

Seven of the 12 major bodies of the Morro Rock–Islay Hill complex were sampled (Figure 2). Sixty-three oriented hand samples were collected in all. Fresh unweathered samples were generally obtainable at recently blasted outcrops, road cuts, and a quarry. At Morro Rock (MR), extensive blasting and quarrying made it possible to collect samples at many points around its perimeter and at various distances inward from the original outer contact. Such a collection made it possible to test for variation of paleomagnetic direction within one of these bodies. As the results will show, there proved to be virtually no variation at all.

PALEOMAGNETISM

Each intrusive body was treated as a single site, and from 5 to 15 oriented hand samples were collected at each site. Two cylindrical specimens drilled from each oriented hand sample

TABLE 1. Paleomagnetism of the Morro Rock–Islay Hill Complex

	N	I, deg	D, deg	k	α_{95}, deg	Remarks*
MR	15	14.6	76.0	80	4.3	af (300 Oe)
BH	8	28.6	39.8	876	1.9	af (330 Oe)
CC	8	45.2	49.8	583	2.3	af (300 Oe)
CP	6	40.6	26.9	51	9.4	af (400 Oe) and thermal
BP	10	38.9	56.1	55	6.6	af (500 Oe)
SLO	5	40.3	33.2	87	8.3	af (300 Oe)
IH	7	70.4	59.3	20	13.8	af (300 Oe) and thermal
Mean of all sites	7	40.9	48.7	14	16.5	
MT†		58.3	−0.31			

N is the number of samples used in site mean computation, I is the site mean inclination, D is the site mean declination, and K and α_{95} are Fisher statistics parameters.

*Magnetic cleaning technique used; values in parentheses are peak af fields used.

†Mid-Tertiary field direction at Morro Bay due to mean mid-Tertiary paleomagnetic pole for stable North America, with coordinates (84.5°N, 115°E), averaged from six paleomagnetic poles in the time interval 25–40 m.y. (M. E. Beck and C. D. Burr, unpublished manuscript, 1978).

were progressively demagnetized in alternating fields to determine an optimum cleaning field for each site. Optimum peak fields ranged from 300 to 500 Oe (Table 1). After cleaning, within site scatter in directions was always low ($k > 50$) except at site IH ($k = 20$). In most cases the paleomagnetic direction after magnetic cleaning differed only slightly from the natural remanent magnetization (NRM) direction. To confirm this stability, several specimens were progressively thermally demagnetized. Again, the magnetization of the specimens proved to be very stable.

Alternating Field Demagnetization

Alternating field demagnetization experiments showed two characteristic types of behavior during progressive demagnetization. Specimens from six of the seven sampled bodies, all except Islay Hill, showed demagnetization characteristics similar to specimen BP6-1 (Figure 3); those from Islay Hill behaved like specimen IH8-1 (Figure 3). In specimen BP6-1, more than three quarters of the NRM intensity was lost at a peak field of 500 Oe (Figure 3*a*). In specimen IH8-1, less than half was lost at 500 Oe, and there was little further decrease upon continued progressive demagnetization to a peak field of 1200 Oe. In both specimens the magnetization direction changed only slightly during demagnetization (Figure 3*b*). Specimens such as BP6-1 have intermediate coercivities, which is indicative of a fine-grained magnetite carrier. Specimens from Islay Hill, such as IH8-1, have both intermediate and high coercivities, indicating that the magnetization probably resides in both fine-grained magnetite and hematite but has the same direction in both.

Thermal Demagnetization

Thirteen specimens, eight from Islay Hill and one from each of five of the other bodies, were thermally demagnetized in four steps (Figure 4). The specimens from Islay Hill retained a large fraction of their NRM intensity after being heated to 640°C, which is above the Curie temperature of magnetite but below the probable blocking temperatures of hematite. Specimen IH8-2 (Figure 4) retained the largest fraction of NRM, almost 80%, indicating that most of its blocking temperatures were above 640°C. All the Islay Hill specimens showed only small changes in their directions of magnetization after heating. After thermal cleaning, the directions were the same, within a few degrees, as those of specimens from the same sample that had been magnetically cleaned.

Of the five specimens from the other bodies, four showed thermal demagnetization characteristics which indicate magnetite as the dominant magnetic mineral. The direction of magnetization of specimen BP9-3, for example, changed only slightly after heating to 550°C (Figure 4*b*); after heating to 640°C, the intensity dropped by an order of magnitude and became random in direction (Figure 4*a*). The fifth specimen, from Chumash Peak, was anomalous in demagnetization behavior compared with other specimens from that site which were demagnetized in alternating fields, showing thermal demagnetization characteristics more like the specimens from

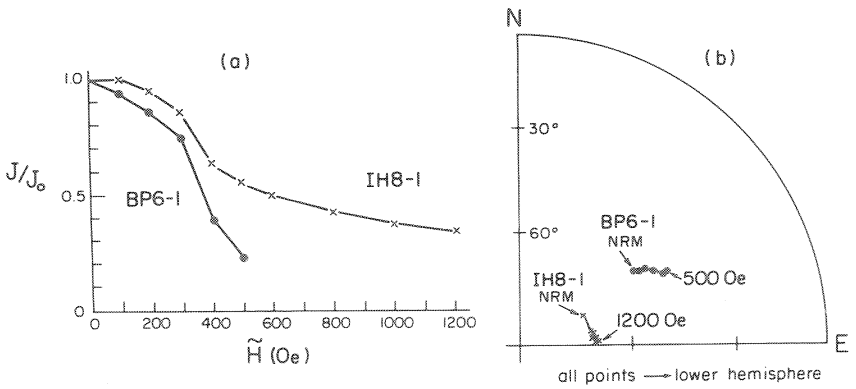


Fig. 3. Alternating field (af) demagnetization of two specimens. Site symbols are the same as in Figure 2. BP6-1 is the first specimen from the sixth oriented sample collected at site BP. (*a*) Decay of NRM intensity with increasing peak demagnetization field. (*b*) Change in magnetization directions of the specimens in part (*a*) during progressive af demagnetization (Schmidt projection).

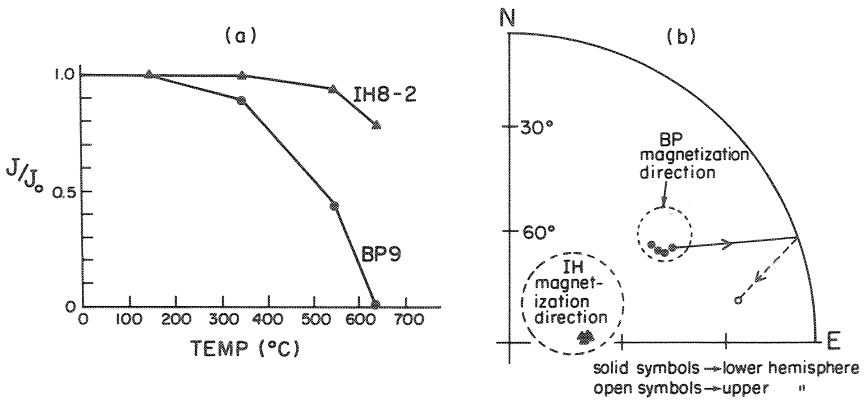


Fig. 4. Thermal demagnetization of two specimens. (*a*) Decay of NRM intensity with increasing temperature. (*b*) Change in magnetization directions of the specimens in part (*a*) during progressive thermal demagnetization. Dashed circles represent 95% confidence limits for the corresponding site means (Schmidt projection).

Islay Hill, with magnetization residing in both hematite and magnetite; but again, the magnetization direction proved to be quite stable.

The thermal and magnetic cleaning experiments demonstrate that the specimens have a single, stable component of magnetization which we conclude was acquired parallel to the geomagnetic field when the intrusives cooled during the Oligocene.

RESULTS

Individual specimen directions were combined to obtain sample means, which were in turn combined to obtain a site mean for each intrusive body. Of the 63 samples collected, 4 were not used because their magnetization directions were highly divergent from those of other samples collected at the same sites, probably owing either to inadvertent sampling of slumped blocks or to untraceable orientation errors. The paleomagnetic directions of all seven sites (Table 1 and Figure 5) are deflected clockwise from the expected mid-Tertiary (MT) field direction by angles ranging from 27° to 76°, the mean being 49°. The inclinations are generally shallower than expected for the mid-Tertiary except at the Islay Hill site, which is steeper. Since this is the only site with hematite as an important magnetic carrier, it is possible that this site has had a more complicated magnetic history than the others.

Superimposed on the 49° clockwise rotation is a considerable amount of scatter in the directions of site means due in part to secular variation in the direction of the ancient paleomagnetic field. The question of whether all of the intersite scatter is due to geomagnetic secular variation or whether part is due to the rotation of sites relative to each other is important in the interpretation that follows. It can be argued that the dispersion of site mean directions is somewhat greater than can be accounted for by geomagnetic secular variation alone. The angular dispersion of the site means has an angular variance $s^2 = 470°^2$. The contribution to this variance made by uncertainty in mean site directions is estimated to be about $20°^2$, so the true between site variance is about $450°^2$. The angular variance expected at this latitude, on the basis of standard models of geomagnetic secular variation [*Cox*, 1970], is about $220°^2$. By assuming a magnetic blocking temperature range $\Delta T_B = 20°$ for these intrusive bodies and using the standard cooling curves of *Lovering* [1935] it is found that the time required for these bodies to cool through ΔT_B is on the order of 10–100 years. This is not a long enough time interval to average out much secular variation, and thus each site mean

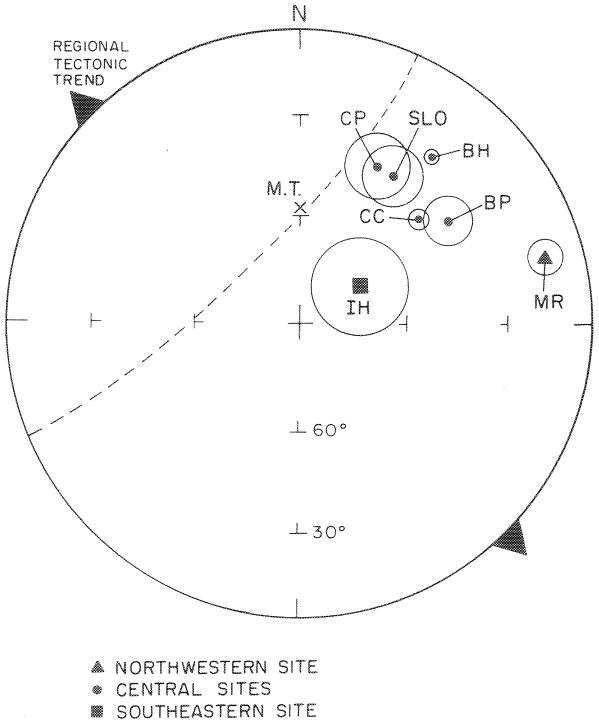


Fig. 5. Site mean directions with circles of 95% confidence for the Morro Rock–Islay Hill complex. The cross is the expected mid-Tertiary (MT) direction for the sampling area. The dashed curve illustrates locations of vectors which would result from rotating MT vector (Table 1) about a horizontal axis parallel to the northwest-southeast regional fold trends (Schmidt projection; all points lower hemisphere).

TABLE 2. Amount of Tilting About a Horizontal Axis Required to Rotate the Expected Mid-Tertiary Field to Coincide With the Observed Paleomagnetic Direction at Each Site

Site	Strike	Dip
MR	N17°E	82°W
BH	N16°W	47°W
CC	N6°E	38°W
CP	N24°W	28°W
BP	N8°E	48°W
SLO	N16°W	32°W
IH	N50°E	28°NW

Amount of tilting is expressed as the present strike and dip at each site of a hypothetical stratum that was horizontal at the time of cooling of each intrusive body.

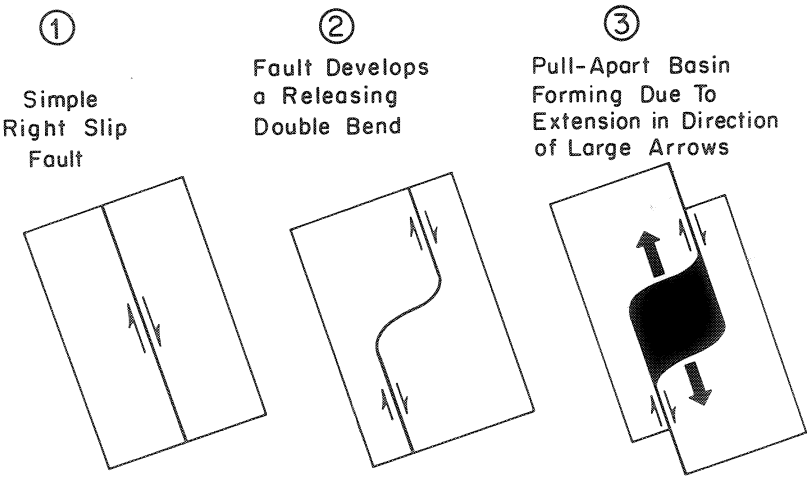


Fig. 6. Formation of a simple pull-apart basin, as described by *Crowell* [1974].

direction should be regarded as an 'instantaneous' reading of the geomagnetic field. Even so, the resulting angular variance of the site means that would be observed if the sites were tectonically undisturbed would be no more than 220°[2]. Therefore part of the observed angular variance of 470°[2] is probably due to tectonic reorientation of each intrusive body. Furthermore, the scatter of site mean directions, aside from site IH, shows much more variation in declination than in inclination. Secular variation, on the other hand, generally produces as much angular change in the direction of increasing and decreasing inclination as in the direction of increasing and decreasing declination. A pattern like that in Figure 5, showing a noticeable streak in declinations, could be generated by starting with a symmetrical angular dispersion of field directions due to geomagnetic secular variation and then rotating some of the field vectors more than others about a vertical axis.

INTERPRETATION

Any set of anomalous paleomagnetic vectors can be accounted for by many different tectonic models, including one of simple tilting about horizontal axes. To investigate whether this makes geologic sense in the present study, we calculated for each site the strike of the local horizontal axis and the amount of tilting about this axis required to produce the observed site direction from the known mid-Tertiary field direction (Table 2). The calculated strikes range from N24°W to N50°E, and the amounts of tilt from 28° to 82°. It is difficult to account for the observed paleomagnetic directions solely by tilting, as the strikes of the required tilt axes are inconsistent with the regional structural trends. The larger faults in this region generally trend in a northwest direction, and nearby post-Oligocene sediments are generally folded about northwest trending axes. If the intrusives had tilted varying amounts about a horizontal northwest axis, all the paleomagnetic directions would lie along the dashed curve in Figure 5. The observed directions, with the possible exception of sites CP and SLO, are well displaced from this curve. It is quite likely that the observed anomalous inclinations are due to modest amounts of tilt about a horizontal axis, as discussed below, but additional large amounts of clockwise rotation about a vertical axis seem to be required to account for the observed paleomagnetic declinations.

Clockwise rotation about a vertical axis is consistent with the known dextral sense of shear along the complex transform boundary in California. One possible rotation mechanism is post-Oligocene penetrative shear distributed throughout the Franciscan melánge and metavolcanics into which these bodies were intruded. The igneous bodies themselves do not appear to have been internally deformed, an interpretation supported by the tight grouping of the paleomagnetic directions of samples collected both near the margins and in the interior of the Morro Rock intrusive. In this sense the intrusive bodies would be rigid 'roller bearings' with nearly vertical axes rotating in a Franciscan matrix which is undergoing penetrative shear.

Alternatively, the Morro Rock–Islay Hill intrusives may

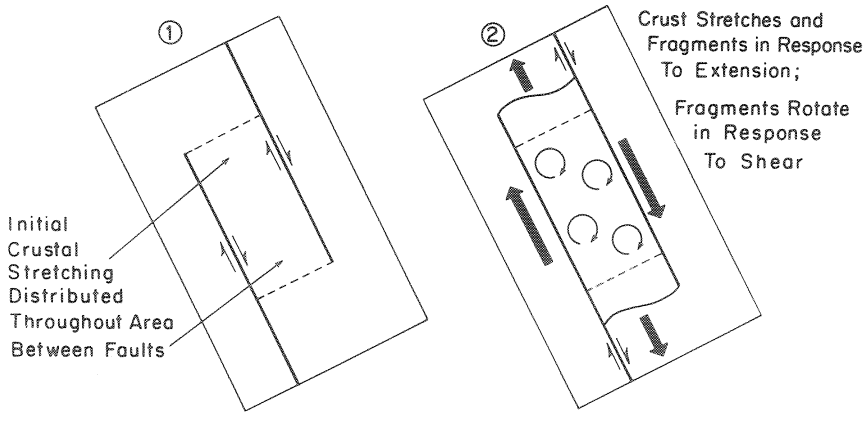


Fig. 7. Modified pull-apart structure showing combined extension and rotation processes.

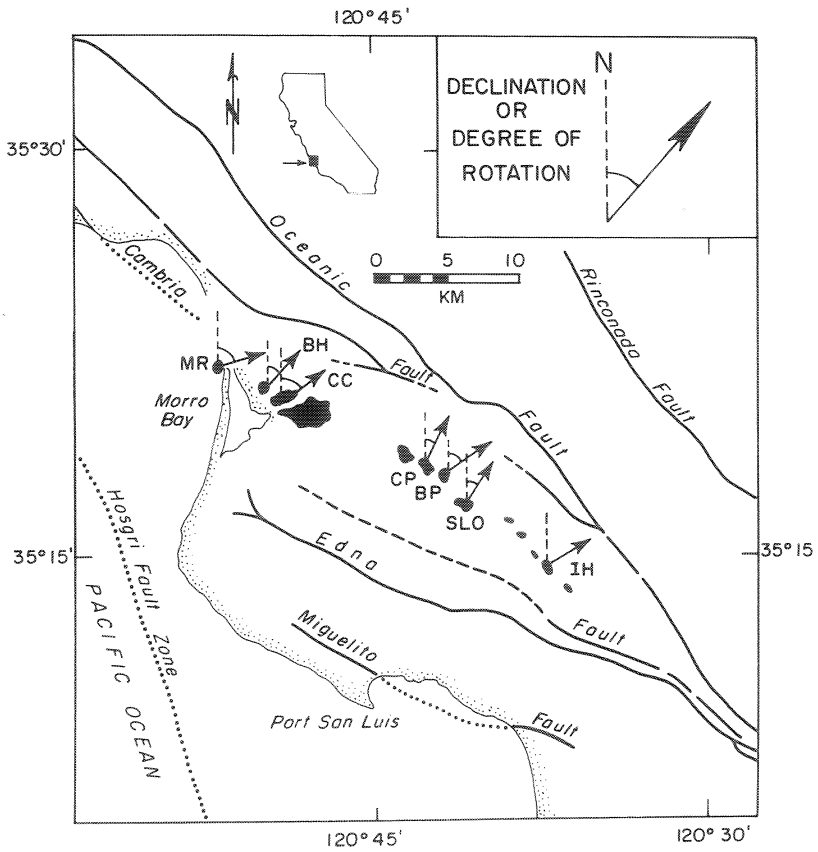


Fig. 8. Directions of horizontal components of magnetization directions of the sampled bodies, Morro Rock–Islay Hill complex. Dashed lines show mid-Tertiary north direction.

have been emplaced during the formation of a pull-apart basin and may have undergone subsequent clockwise rotations along with extensionally loosened small crustal blocks within the basin. The idea of a pull-apart basin as an important tectonic feature in southern California was first developed by *Crowell* [1974]. If a strike slip fault develops an appropriate double bend (Figure 6), continued right slip will produce extension and subsidence as the adjacent blocks pull apart, resulting in the formation of a basin. With continued extension the basin floor will rupture, allowing the emplacement of volcanics at the center of the basin. C. A. Hall (unpublished manuscript, 1977) previously suggested that the Morro Bay basin may have developed as a pull-apart structure, on the basis of general similarities to another proposed pull-apart basin, the Santa Maria–Lompoc basin to the south.

In applying this idea to the Morro Bay area the following modification seems appropriate. The pull-apart structure is viewed as having begun with overlapping en echelon right slip faults, rather than as having formed from a fault with a double bend, as is shown in Figure 7. This fault pattern is observed quite commonly today in the San Andreas transform zone. With this geometry, initial extension would be distributed throughout the entire length of the long, narrow area between the overlapping faults (Figure 7 (left)), which would account for the elongate pattern of emplacement displayed by the Morro Rock–Islay Hill intrusives. In a simple pull-apart basin the length of the basin is equal to the amount of strike slip displacement since the time of formation of the double bend in the original strike slip fault. In a pull-apart structure such as is shown in Figure 7 the length of the basin is determined by the initial amount of overlap of the en echelon faults plus the amount of subsequent strike slip displacement; depression of the basin floor is less pronounced because extension is distributed throughout the length of the fault overlap, and volcanism is discontinuous and occupies only a small fraction of the area of the basin, as is observed in the Morro Bay area.

After initial crustal fragmentation and volcanism, continued right slip on the faults and accompanying extension in the basin would loosen the crustal fragments, or blocks, and allow them to rotate clockwise in response both to dextral shear exerted by the bounding faults and to distributed shear flow in the underlying ductile part of the crust (Figure 7 (right)). The crustal blocks would undergo different amounts of rotation if, as seems likely, the crust was originally heterogeneous and the initial pattern of crustal fragmentation was irregular. The paleomagnetic results suggest that differential rotation has, in fact, occurred between at least three blocks. The different paleomagnetic directions for the five sites in the central part of the chain of intrusives (Figures 5 and 8) have an angular variance of 150°[2], which could easily be due to geomagnetic secular variation, although the possibility of small rotations between these sites cannot be ruled out on paleomagnetic grounds. However, the paleomagnetic directions for site MR at the northwest end of the chain and site IH at the southeast end are displaced by large angles of 35° and 30° from the mean direction of the central group of sites and are displaced by 57° from each other. If, as seems likely, differential rotation has occurred between sites, the paleomagnetic directions suggest that perhaps site MR at the northwest end of the chain, site IH at the southeast end, and the central group of five sites are on three differentially rotated crustal blocks.

The paleomagnetic results offer some hints about the sizes of

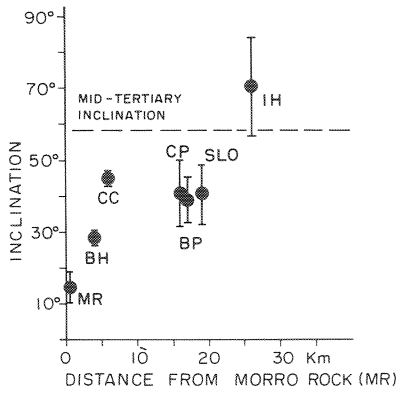


Fig. 9. Paleomagnetic inclination as a function of position along the chain of intrusives.

the crustal blocks which have rotated coherently with little internal deformation. The sites of the central block (BH, CC, CP, BP, and SLO) span about 20 km, a length that is similar in magnitude to the spacing of major faults in this region. It is also roughly equal to the maximum focal depth of earthquakes along the San Andreas fault zone, the absence of earthquakes at greater depth being attributed to a transition from brittle to ductile deformation. Twenty kilometers would appear to be a reasonable diameter for coherent crustal blocks decoupled from a ductile lower lithosphere.

At all but one of the sites the paleomagnetic inclination is shallower than expected for the mid-Tertiary field. If no tilting has taken place about horizontal axes, the difference of 17° between the mid-Tertiary inclination and the observed mean inclination (Table 1) would correspond to a northward translation of 1700 km for this crustal block. An alternative explanation is that some minor tilting has taken place. This would not be unexpected in a pull-apart basin, as tilting of crustal blocks commonly accompanies normal faulting in regions of crustal extension. The observed paleomagnetic inclinations (and therefore the inferred amounts of tilt) are spatially correlated along the length of the chain of intrusives, with shallower inclinations to the northwest and steeper inclinations to the southeast (Figure 9). This spatial coherency supports the interpretation that the angular dispersion in the paleomagnetic field directions is not due entirely to geomagnetic secular variation but rather reflects some spatially coherent tectonic process. For example, the observed paleomagnetic inclinations would be consistent with tilting of both the northwest and the southeast blocks inward toward the central block, as might be expected in a pull-apart basin with a structural low in the middle and sloping ramps on either end. Additional tilting of the central block toward the south would be required to account for the shallow inclinations of sites BH, CC, CP, BP, and SLO.

Discussion

Pull-apart basins are but one type of crustal response to regional shear. *Freund* [1974] suggested other ways in which crustal blocks might rotate within a sheared region. The broad transform zone between the Pacific and North American plates, with its offset transform faults, seems to provide an environment favorable for the combined pull-apart and rotation process. Evidence which suggests a direct relationship between the shearing of coastal California and the proposed evolution of the Morro Bay area would be supportive of the ideas presented in this paper. *Atwater*'s [1970] reconstruction of plate tectonic activity along the west coast of North America indicates that at any point along the California coast, shearing began with the passing of the Mendocino triple junction, as its northward migration marked the transition from a subductive plate boundary to a transform boundary. Figure 10 shows past positions of the Mendocino triple junction along the edge of the North American plate, based on reconstructions by *Atwater and Molnar* [1973] and *Graham and Dickinson* [1978]. The large arrows indicate past positions of the triple junction with respect to the displaced continental sliver between the San Andreas fault and the San Gregorio–Hosgri fault, which contains the Morro Rock–Islay Hill complex. The range of radiometric ages obtained for the complex, 22.1 ± 0.9 m.y. to 26.5 ± 0.8 m.y., indicates that these bodies were intruded just after the Mendocino triple junction passed this area [*Dickinson and Snyder*, 1979]. This suggests a direct relationship between this volcanic activity and the beginning of strike slip faulting in this area, which would be consistent with the proposed model. The simple shear of the faults could not have directly produced the volcanism but may have produced the extensional pull-apart process which, in turn, resulted in volcanism.

Conclusions

The present study supports *Freund*'s [1974] suggestion that crustal blocks in sheared regions have commonly undergone rotation and *Beck*'s [1976] hypothesis that dextral shear and clockwise rotation are characteristic of much of the tectonism in western North America. Whereas in previous studies, rotated crustal blocks were found to have dimensions of several hundred kilometers and could legitimately be regarded as microplates, in the Morro Bay area the rotated blocks have dimensions of 20 km or less and may well be fragments of the

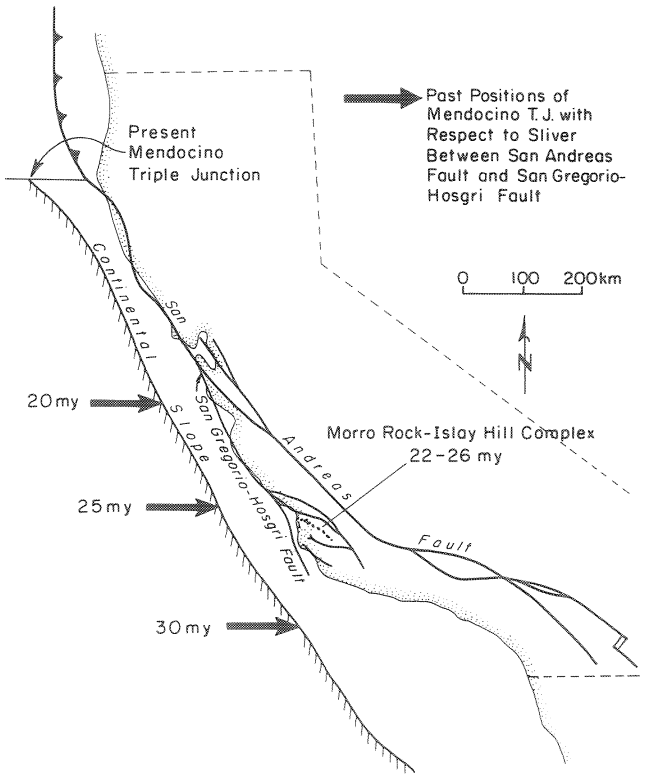


Fig. 10. Diagram of the western edge of North America showing past positions of the Mendocino triple junction and the location and age of the Morro Rock–Islay Hill complex (modified from *Atwater and Molnar* [1973], *Dickinson and Snyder* [1979]).

brittle upper part of the crust and not separate segments of lithosphere. As most of coastal California exhibits a broad zone of parallel, en echelon, and branching dextral faults, the combined pull-apart and rotation process may have occurred in many areas within the region. Although volcanism may not always occur as part of the process, there may be numerous localities where small extensionally loosened crustal blocks have been rotated.

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